

Speech Based Parkinson's Disease Detection Using LPC Coefficients and Residual Energy

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Abstract—Detecting Parkinson's Disease (PD) using speech signals is an important step toward early diagnosis and continuous health monitoring. In this project, we developed a machine learning-based system that analyzes speech using Linear Predictive Coding (LPC) and residual energy features. These features are then used with an XGBoost classifier to distinguish between healthy individuals and patients with Parkinson's.

The speech signals are first preprocessed (silence removal and normalization), then divided into frames. From each frame, LPC coefficients and residual energy are extracted, and statistical measures like mean, standard deviation, maximum, and minimum are calculated. This results in a 56-dimensional feature vector for each audio sample.

The model was tested on a dataset of 80 speech recordings. It achieved an average accuracy of 66.25

Index Terms—Parkinson's Disease, Linear Predictive Coding (LPC), Residual Energy, XGBoost, Speech Signal Analysis, Feature Extraction.

I. INTRODUCTION

Parkinson's disease (PD) is a progressive neurological disorder that affects the nervous system and the parts of the body controlled by nerves. One of the earliest and most common symptoms of PD is the degradation of speech and vocal capabilities, often manifesting as reduced volume, breathiness, and impaired articulation (hypokinetic dysarthria). The automatic identification of these vocal biomarkers from speech signals has gained significant importance as a non-invasive diagnostic tool.

Over the years, several approaches have been proposed for PD detection, many of which rely on complex deep learning models like Convolutional Neural Networks (CNN) or Transformers. Although these methods show promising results, they often require massive datasets and high computational resources. In contrast, classical acoustic features combined with robust ensemble learning algorithms can provide a highly interpretable and computationally efficient alternative.

This work adopts a feature extraction pipeline utilizing Linear Predictive Coding (LPC). LPC accurately models the vocal tract as a filter, capturing the resonant frequencies (formants) of speech. In addition, residual energy features are extracted to characterize the glottal excitation and signal complexity. An XGBoost classifier is then employed to evaluate the effectiveness of the proposed statistical feature set.

II. PROPOSED METHOD

A. Dataset

The proposed PD detection system was evaluated using a customized dataset of speech recordings containing both pathological and control subjects. After removing corrupted files, the final dataset yielded 80 audio samples, comprising 41 Healthy Controls (HC) and 39 Parkinson's Disease (PD) patients. The dataset was split into an 80:20 training and testing ratio, ensuring stratified class representation.

B. Preprocessing and Theoretical Foundations

The proposed system begins with the preprocessing of the input speech signal to remove silent artifacts and standardize the acoustic data.

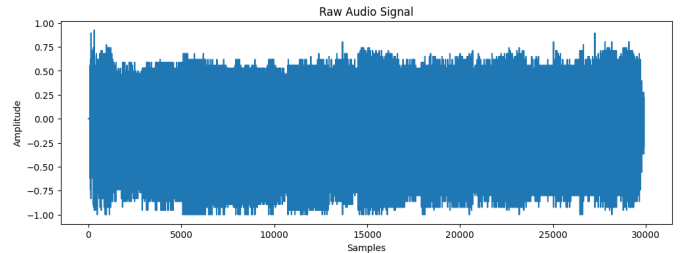


Fig. 1. Raw Audio Signal from the dataset.

1) *Silence Removal and Normalization*: The leading and trailing silences are trimmed using a 20 dB threshold to ensure only active speech is analyzed. The signal is then amplitude-normalized to reduce variations caused by different recording equipment.

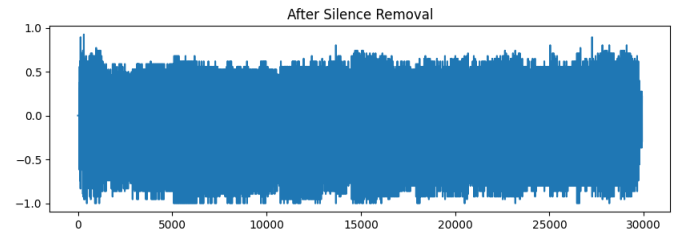


Fig. 2. Audio Signal after Silence Removal.

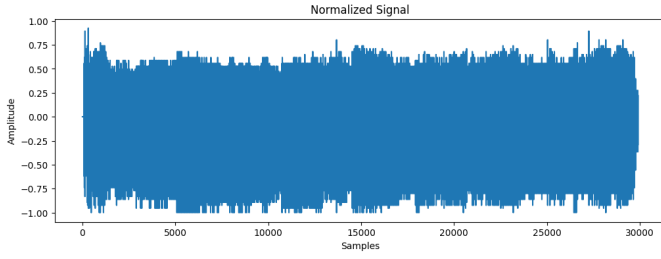


Fig. 3. Normalized Audio Signal ensuring amplitude consistency.

2) *Pre-emphasis Filtering*: Speech signals naturally exhibit a spectral roll-off, where higher frequencies have less energy than lower frequencies. To balance the frequency spectrum and improve the numerical stability of the LPC analysis, a first-order high-pass pre-emphasis filter is applied:

$$y[n] = x[n] - \alpha x[n-1] \quad (1)$$

where $\alpha = 0.97$ is the pre-emphasis coefficient.

3) *Framing and Windowing*: Because speech is a quasi-stationary signal, it is segmented into short overlapping frames (20 ms frame length, 10 ms hop length). A Hamming window is applied to each frame to minimize spectral leakage at the boundaries:

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right) \quad (2)$$

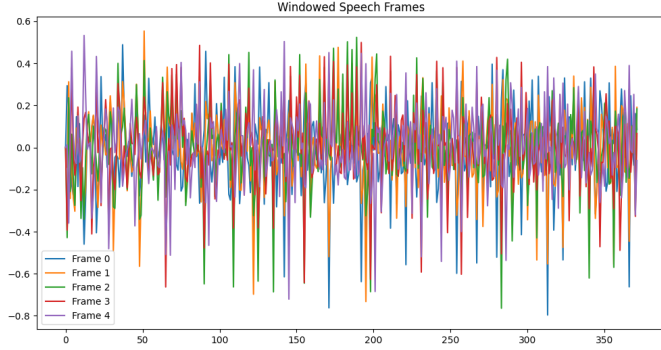


Fig. 4. Windowed Speech Frames prepared for feature extraction.

C. LPC and Energy Feature Extraction

1) *Linear Predictive Coding (LPC)*: LPC is based on the source-filter model of speech production, which assumes that the human vocal tract acts as an Infinite Impulse Response (IIR) filter. The core idea is that the current speech sample $s[n]$ can be approximated as a linear combination of p past samples:

$$\hat{s}[n] = \sum_{k=1}^p a_k s[n-k] \quad (3)$$

where a_k are the LPC coefficients and p is the predictor order. In this study, we utilize an LPC order of $p = 12$, which is optimal for capturing the vocal formants of human speech.

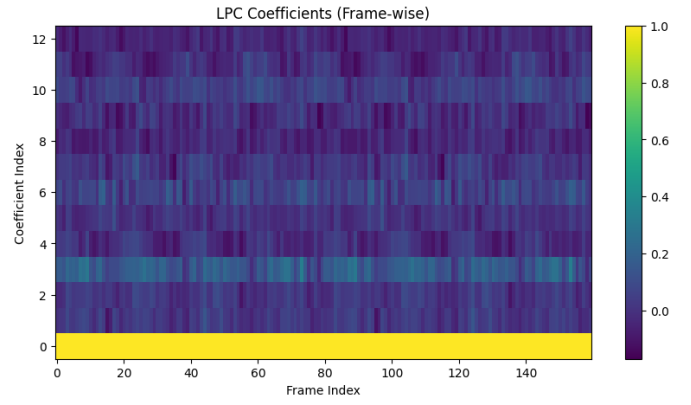


Fig. 5. LPC Coefficients mapped frame-wise.

2) *Residual Signal and Energy*: The difference between the actual speech sample and the predicted sample is the prediction error, or the *residual signal* $e[n]$:

$$e[n] = s[n] - \hat{s}[n] \quad (4)$$

The residual signal contains the information regarding the glottal source excitation (vocal cord vibrations).

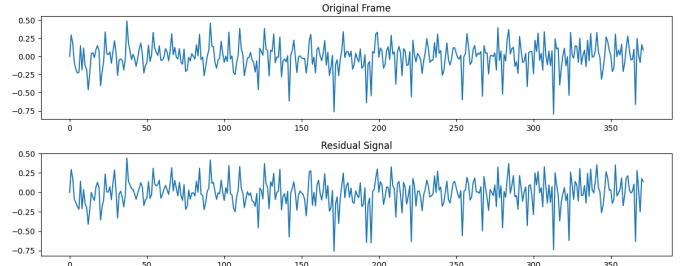


Fig. 6. Comparison of an Original Frame and its Residual Signal.

The residual energy for each frame is calculated to measure the intensity of the glottal excitation:

$$E_{res} = \sum_n e[n]^2 \quad (5)$$

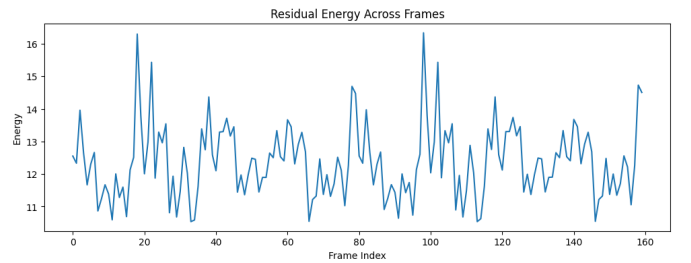


Fig. 7. Residual Energy calculated across all frames.

3) *Feature Aggregation*: To create a fixed-length feature vector for the classifier, statistical aggregations (Mean, Standard Deviation, Maximum, Minimum) are performed across all frames for a single audio file. Applying these 4 statistics

to the 13 LPC coefficients (including the zeroth coefficient) and 1 residual energy yields a 56-dimensional feature vector per audio sample.

D. XGBoost Classification

Extreme Gradient Boosting (XGBoost) is an advanced implementation of gradient boosted decision trees. It minimizes a regularized objective function to prevent overfitting:

$$\mathcal{L}(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k) \quad (6)$$

where l is a differentiable convex loss function that measures the difference between the prediction $\hat{y}_i$ and the target y_i , and Ω penalizes the complexity of the trees. The model was configured with 200 estimators, a maximum depth of 4, and a learning rate of 0.05.

III. RESULTS AND DISCUSSION

The performance of the proposed acoustic pipeline was evaluated using a 5-fold cross-validation on the dataset, followed evaluation on the 20% unseen test set.

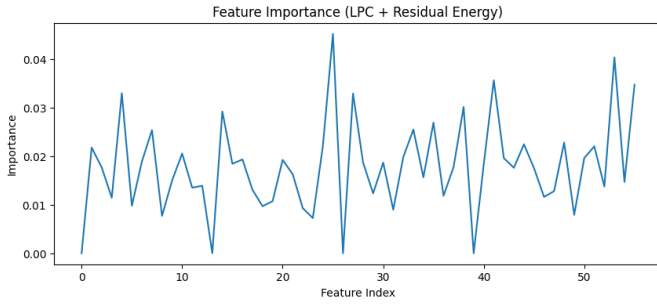


Fig. 8. Feature Importance of the 56-dimensional vector determined by the XGBoost algorithm.

A. Cross-Validation Performance

To ensure the stability and reliability of the model across different data subsets, a 5-fold cross-validation was executed. As shown in Table I, the model achieved a mean accuracy of 66.25%.

TABLE I
5-FOLD CROSS-VALIDATION SCORES

Fold	Accuracy (%)
Fold 1	68.75
Fold 2	62.50
Fold 3	50.00
Fold 4	75.00
Fold 5	75.00
Mean Accuracy	66.25%

The variance across folds indicates that the model's performance is sensitive to the specific train-test splits, which is a common characteristic when working with smaller biological datasets.

B. Probability and Threshold Analytics

Further analysis of the model's predictive probabilities is visualized using the Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves.

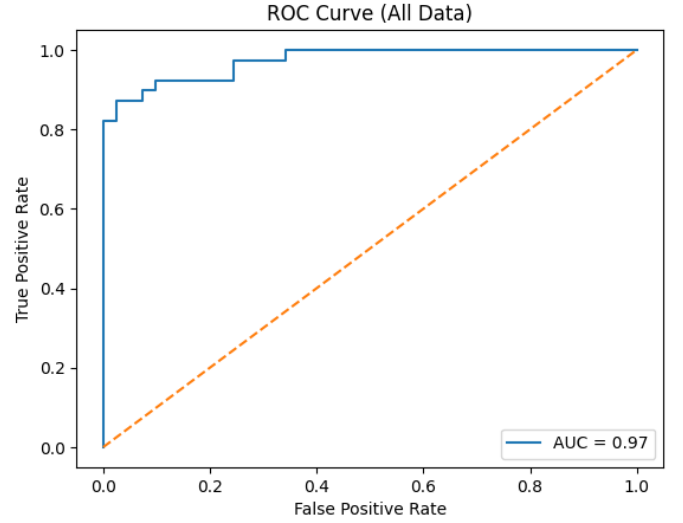


Fig. 9. Receiver Operating Characteristic (ROC) Curve evaluated across the dataset (AUC = 0.97).

As shown in Fig. 9, the ROC curve plots the True Positive Rate against the False Positive Rate, achieving an Area Under the Curve (AUC) of 0.97 when evaluated across the entire combined dataset.

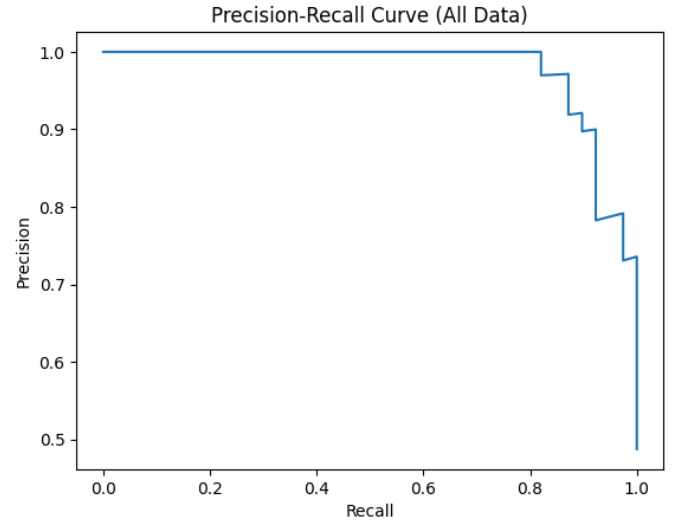


Fig. 10. Precision-Recall Curve illustrating the trade-off between precision and recall.

Fig. 10 displays the PR curve, highlighting the trade-off between precision and recall at various threshold settings. While these high AUC bounds demonstrate the model's strong theoretical capacity to separate classes, they also reflect the optimistic bias of evaluating on the combined (train + test) data, reinforcing the necessity of strict hold-out evaluations.

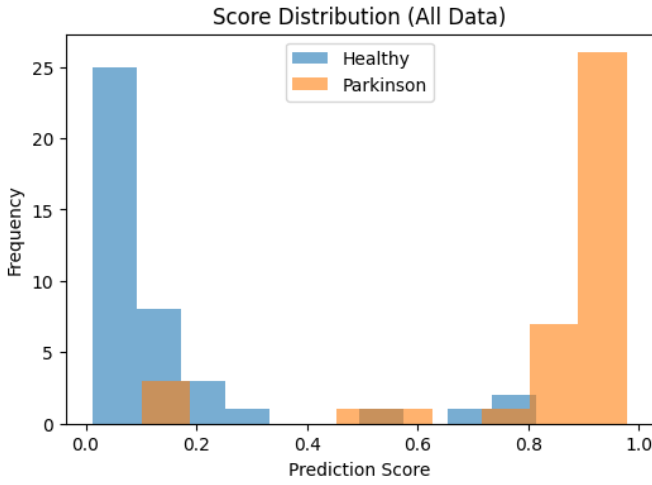


Fig. 11. Score Distribution separating Healthy and Parkinson's predictions.

C. Test Set Evaluation

On the strictly partitioned 20% hold-out test set (8 Healthy, 8 Parkinson's), the XGBoost classifier achieved an overall accuracy of 56.25%. A detailed breakdown of the model's predictive performance is presented in Table II.

TABLE II
CLASSIFICATION REPORT (TEST SET)

Class	Precision	Recall	F1-Score	Support
Healthy (0)	0.57	0.50	0.53	8
Parkinson (1)	0.56	0.62	0.59	8
Macro Avg	0.56	0.56	0.56	16
Weighted Avg	0.56	0.56	0.56	16

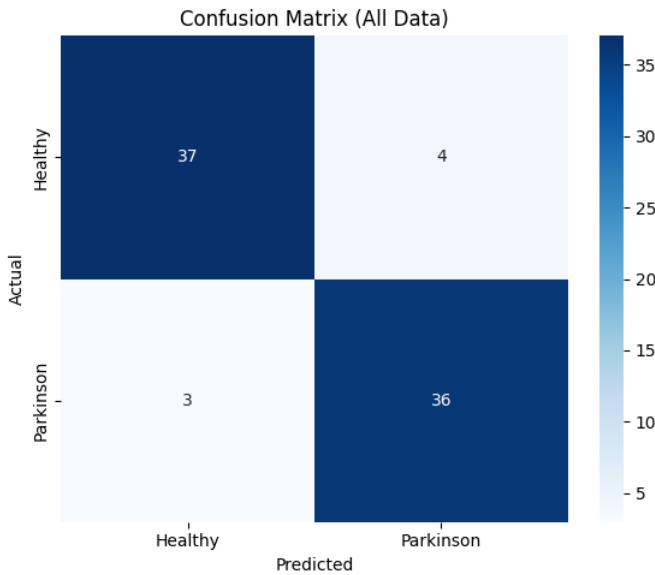


Fig. 12. Confusion Matrix representing the classification performance.

The classification report and confusion matrix reveal that the model is slightly better at recalling Parkinson's patients (Recall = 0.62) than Healthy patients (Recall = 0.50), reflecting a slight threshold bias toward predicting the positive class.

IV. CONCLUSION

In this work, a Parkinson's Disease detection system based on Linear Predictive Coding (LPC), residual energy features, and an XGBoost classifier was presented. By aggregating statistical measures across speech frames, a 56-dimensional feature vector was successfully extracted for 80 subjects.

The model achieved a peak cross-validation accuracy of 75.00% (with a mean of 66.25%) and a strict test set accuracy of 56.25%. While the pipeline successfully discriminates between classes better than random chance, the results indicate that the overlapping acoustic boundaries between healthy and pathological speech require larger datasets to generalize effectively. Future work will focus on expanding the dataset size, tuning hyperparameters, and exploring the fusion of LPC with non-linear entropy features.

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